

Welcome to

Data Science



INF-M
Winter Term 2026/27

Prof. Dr. Markus Breunig



Lecture 1

—

Introduction

Amsel / Common Blackbird (*Turdus merula*)

- ◆ The common blackbird breeds in Europe, Asiatic Russia, and North Africa, and has been introduced to Australia and New Zealand and is the most common bird in Germany. Depending on latitude, the common blackbird may be resident, partially migratory, or fully migratory.
- ◆ The adult male of the common blackbird, which is found throughout most of Europe, is all black except for a yellow eye-ring and bill and has a rich, melodious song. The adult female & juvenile have mainly dark brown plumage.
- ◆ They breed in woods & gardens, building a neat, cup-shaped nest, bound together with mud. They are omnivorous, eating a wide range of insects, earthworms, berries, and fruits.
- ◆ Anyone who observes blackbirds closely will notice that the animals very often hop on the ground, pause briefly and tilt their heads: they listen to where earthworms, snails and insects are crawling in the ground.

Sources:

* Photo by MabelAmber on pixabay

* https://en.wikipedia.org/wiki/Common_blackbird

* <https://www.br.de/radio/bayern1/heimische-vogelarten-100.html>

Focusing Objectives

By the end of today's lecture, you will be able to:

1. Understand the Role of Data Science
2. Apply Data-Driven Decision-Making (DDDM)
3. Recognize and Describe the CRISP-DM Process
4. Familiarize yourself with the Technical Ecosystem



Data Science

Introduction

1. Class Setup
2. What is Data Science (compared to ML/AI/...)
3. DDDM - from Descriptive to Prescriptive Analytics
4. The CRISP-DM Process
5. Tech Stack: Python and Jupyter Notebooks

Class Setup

◆ Schedule

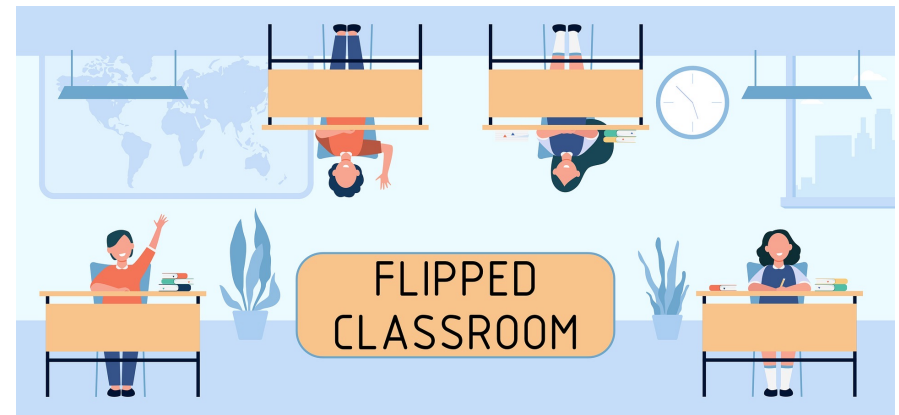
- 4 hours class time per week (2 hours lecture + 2 hours exercises)
- Homework assignments – to be completed before the respective class

◆ Flipped (inverted) classroom concept

- Basics of new concepts introduced before lectures (in the homework assignments)
 - Classes consist of advanced lectures and exercises/discussions
 - See https://en.wikipedia.org/wiki/Flipped_classroom
- Homework assignment are mandatory and critical

◆ Exam

- Oral Exam, 15 min
No aids allowed



Learning Goals and Skills

◆ Learning Goals

- The participants know and understand the methodological approach and theoretical basics of Data Science and can solve real Data Analytics problems.

◆ Skill to be improved in this class

- Fundamentals of Data Science: methods & tools to solve analytical and predictive problems
→ The theory and processes of data science
- How to approach real-world Data Science problems with widely used (esp. open-source) tools
→ The practical implementation of data science
- Time management and self-directed learning and problem solving
- Teamwork and discussion skills

Contents of the class and Prerequisites needed

◆ Content of this class

- Data Science Process – CRISP-DM
- Basics of Data Science – Business problem vs. ML problem
- EDA (exploratory data analysis) and data preparation
- Feature engineering, Modelling, Tuning and Evaluation
- Useful Python libraries for Data Science (pandas, matplotlib, seaborn, scikit-learn)
- Evaluating results from the Business Perspective
- Visualising and communicating data and results

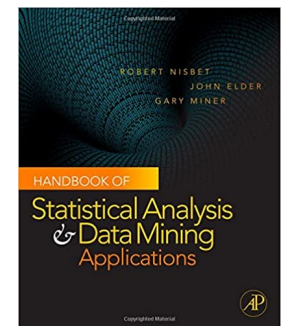
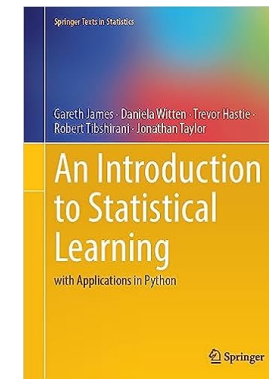
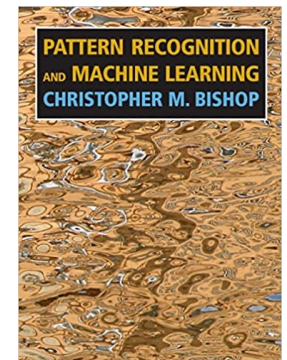
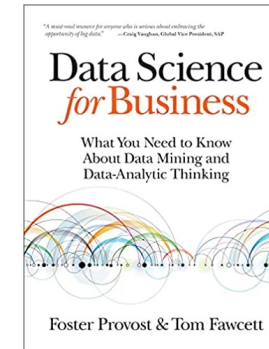
◆ Prerequisites

- Relational databases and SQL
- Machine Learning Algorithms:
Supervised Learning, Unsupervised Learning
- Python and Object-oriented programming incl. NumPy
- Statistics (basic know-how)
- ...and a curious mind!



Literature

- ◆ Foster Provost, Tom Fawcett:
Data Science for Business.
(O'Reilly, 2013)
- ◆ Christopher M. Bishop:
Pattern Recognition and Machine Learning.
(Springer, 2006)
- ◆ Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani:
An Introduction to Statistical Learning (with appl. in Python).
(Springer, 1st Edition, 2023)
- ◆ Robert Nisbet, John Elder IV, Gary D. Miner:
Handbook of Statistical Analysis and Data Mining Applications.
(Academic Press, 2009)



...so what's up with the Birds?

- ◆ State of the World's Birds report 2022
 - nearly **half of all bird species are in decline**
 - **one in eight at risk of extinction**
 - pressures causing these declines are well understood
 - vast majority are driven by **human actions**.
- ◆ Learn about the **most common birds in Germany**
- ◆ Take the time to pause when you see a bird and **appreciate their beauty** while you still can



Photo by Ton W on Pixabay
Source: <https://www.birdlife.org/papers-reports/state-of-the-worlds-birds-2022/>

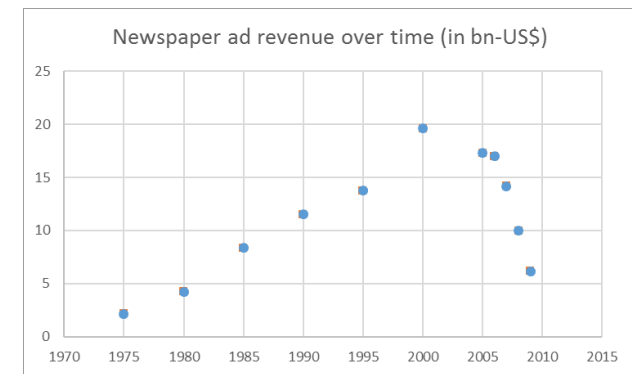
Data Science

Introduction

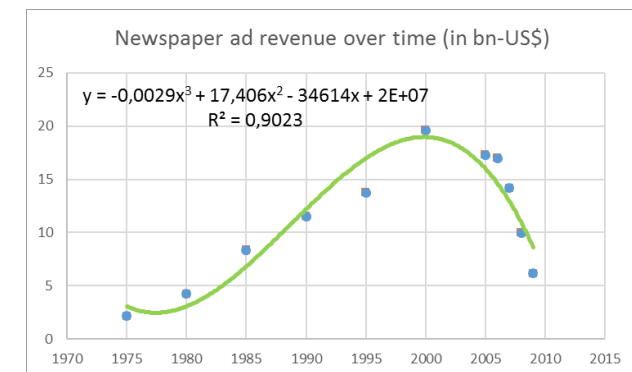
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2. What is Data Science (compared to ML/AI/...)
3. DDDM - from Descriptive to Prescriptive Analytics
4. The CRISP-DM Process
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What is Machine Learning?

- ◆ Computer solves a problem without an explicitly coded algorithm
- ◆ Instead: use a highly parameterized algorithm and “learn” from data
- ◆ Model = Set of parameter values
- ◆ Compute the parameter values (the model) using a training algorithm and (lots of) examples (data)
- ◆ Most ML methods employ 2 algorithms
 - one for training the model and
 - one for model execution/evaluation



original dataset



polynomial regression of degree 3

Why use Machine Learning?

- ◆ For many real-world tasks, it is **difficult to come up with an explicit** (i.e. rule-based) **algorithm**
 - Too many rules needed to cover all the cases
 - Many exceptions (including exceptions to exceptions!) to be handled
 - Expert knowledge needed (i.e., an expert to handcraft the rules)
 - Rules can be difficult to
 - Design – rules interact in unpredictable ways
 - Maintain – adding new rules can easily break everything
 - Adopt to new domains – we need to significantly modify/add rules
 - Many tasks are often inherently subjective, it is difficult to model subjectivity with rules
- ◆ It is often **easier to manually label data** than to design an explicit algorithm
 - Labelling typically requires less expert knowledge
 - We don't care how complex or subjective the task is
 - We let the data „speak for itself” and machine learning algorithm to do the work
 - Labelled data may already be readily available



Photo by Austin Distel on Unsplash

Kinds of machine learning

Supervised learning

- Labelled data points (class or value)
- Goal: study many labelled examples, make predictions about future data points
- **Tasks:** classification, regression, link prediction/recommender, ...

Unsupervised learning

- Data points are not labelled
- Goal: organize the data in some way or describe its structure
- **Tasks:** clustering, anomaly detection, dimensionality reduction, association rules, topic-modelling, ...

Semi-Supervised learning

- Small portion of data points are labelled, large number of data points are unlabelled
- Methods: self-training, co-training, specific algorithms...
- **Tasks:** all supervised tasks, some unsupervised tasks (like anomaly detection)...

Reinforcement learning

- Choose an action in response to a data point, receive reward signal
- Goal: modifies the response strategy in order to achieve the highest reward

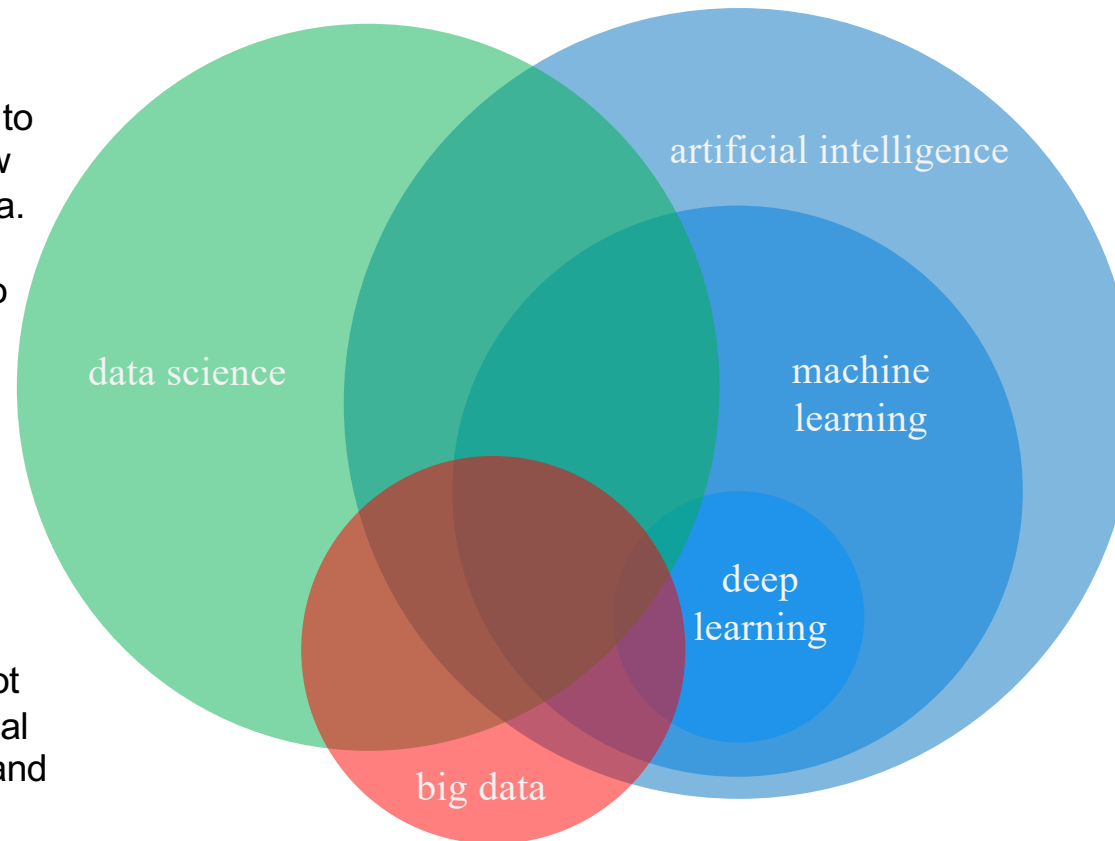
AI vs. ML vs. DL

Data Science

Uses scientific methods, processes, and algorithms to obtain knowledge and draw insights from observed data. Goal does not have to be model training, often aim to extract knowledge and insight to support human decision-making without creating an AI/ML system.

Big Data

Efficiently analyse large amounts of data that cannot be processed with traditional data-processing methods and applications.



Artificial Intelligence (AI)

Techniques, that enable computers to solve tasks similar to the way humans do

Machine Learning (ML)

(Mostly statistical) Algorithms, that allow computers to learn from data (examples, experiences) without being explicitly programmed

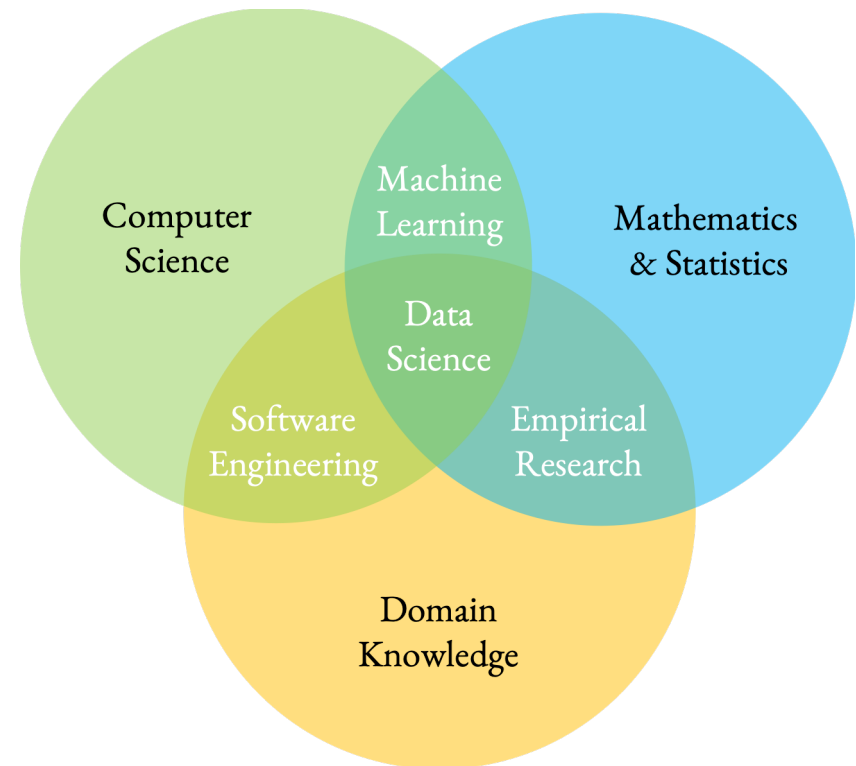
Deep Learning (DL)

A subset of ML which uses multi-layer (deep) neural networks (often using vast amounts of data with little or no pre-processing)

What is Data Science?

“**Data Science** is an interdisciplinary field about processes and systems to **extract knowledge or insights from data** in various forms, either structured or unstructured which is a continuation of some of the data analysis fields such as statistics, machine learning, data mining, and predictive analytics, similar to Knowledge Discovery in Databases (KDD).”

Goal of data science: transform raw data into actionable insights that can inform business decisions (and drive innovation).



Source: https://en.wikipedia.org/wiki/Data_science

Why should I care?

- ◆ It's lots of **fun**!
- ◆ Harvard Business Review named
 - Data Scientist the **The Sexiest Job of the 21st Century**⁽¹⁾
- ◆ Data Science Know-How is a **valuable skillset in high demand** (in Germany und world-wide):
 - "Data Scientist gehören zu den gefragtesten Experten am Arbeitsmarkt. Das Statistik-Portal Statista schätzt einen Bedarf von mehr als vier Millionen Fachkräfte für 2015."⁽²⁾
 - "Data Scientists – Die begehrtesten Alleskönner des 21. Jahrhunderts"⁽³⁾
 - "The median salary of a junior level data scientist is \$91.000, but those managing a team of ten or more data scientists earn base salaries of well over \$250,000, according to Burtch Works. Supply is still tight ..."⁽⁴⁾

IT-Jobs nach Brutto-Mediangehalt	
Job	Brutto-Mediangehalt pro Jahr
Data Scientist	67.000
Product Owner/in	63.500
Data Engineer	60.750
Business-Analyst/in	60.500
Controller/in	59.500
Software Engineer	57.500
Data Analyst/in	56.750
Software-Entwickler/in	54.500
Wirtschaftsinformatiker/in	52.500
Quelle: Stepstone	

(5)

Sources:

(1) HBR: <https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century>

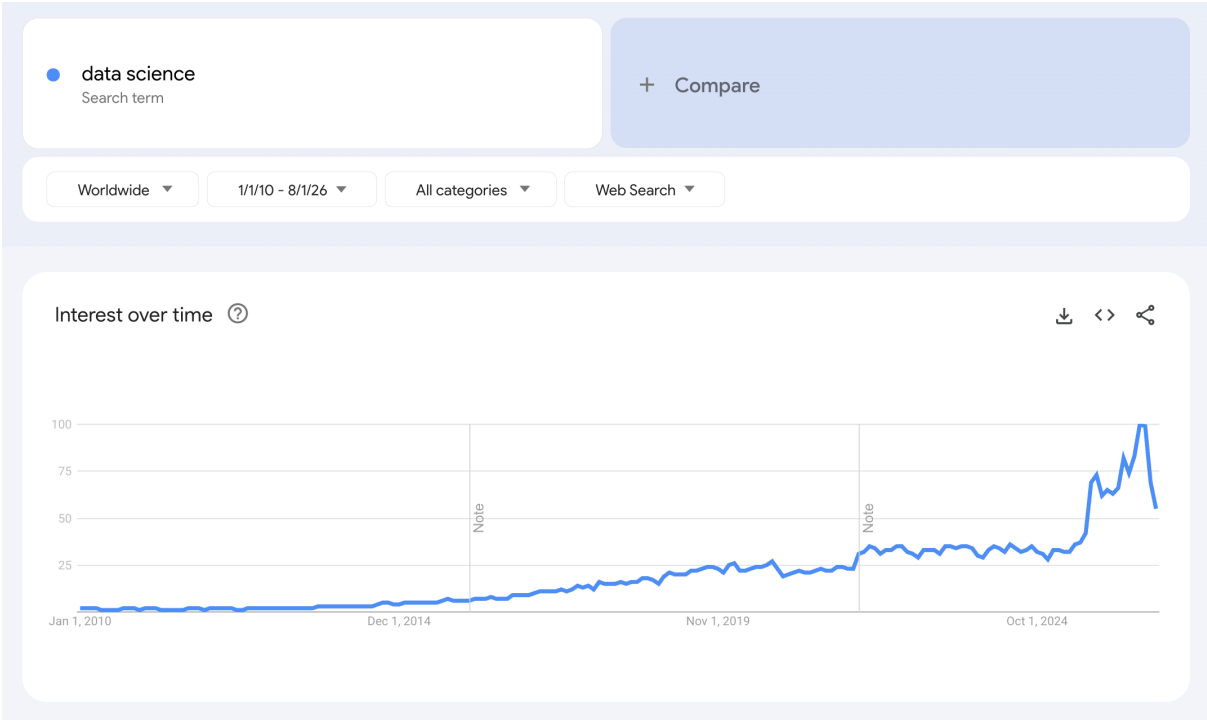
(2) Der Standard: <http://derstandard.at/2000012040062/Grosse-Nachfrage-nach-Data-Scientists>

(3) Capgemini Consulting: <https://www.de.capgemini-consulting.com/blog/digital-transformation-blog/2014/01/data-scientists-die-begehrtesten-alleskonner-des-21>

(4) Forbes, <http://www.forbes.com/sites/gilpress/2015/04/30/the-supply-and-demand-of-data-scientists-what-the-surveys-say/#6dadf1c4205e>

(5) <https://www.heise.de/news/Jobmarkt-Was-KI-Experten-an-Vergeltung-erwarten-koennen-9628584.html>, February 2024

Data Science – Google Trends



Gartner.

Gartner Hype Cycle for Emerging Technologies, 2014



Sources:
(1) <https://trends.google.com/trends/explore?date=2010-01-01%202026-08-01&q=data%20science&hl=en&legacy>
(2) <https://www.forbes.com/sites/gartnergroup/2014/09/17/digital-business-technologies-dominate-gartner-2014-emerging-technologies-hype-cycle/>

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The Old Way – Intuitive Decision Making

- ♦ Traditionally: intuition (=simply “knowing” when something is right or wrong) → tremendous amount of prestige, importance, and influence.
 - 50.3 percent of all Americans agreed with the statement “I trust my gut to tell me what’s true and what’s not.” (source: survey in 2016 by K. Garret, B. Weeks; <https://theconversation.com/should-we-worry-that-half-of-americans-trust-their-gut-to-tell-them-whats-true-84259>)
- ♦ But: **Intuition is frequently wrong!**
 - Humans are bad at interpreting various situations and can easily be misled
 - See e.g. D. Kahnemann (nobel prize winner): “**Thinking, Fast and Slow**” (2012, ISBN 978-0141033570)

→ millions of Americans believe things that aren’t true

Intuitive Decision Making



- Relies on Gut Feeling (Subjective)
- Relies heavily on Previous Experience
- Best Guesses: Inherently Risky
- Correctiv & Unpredictable

The Better Way – Data Analytic Thinking and Data Driven Decision Making

- ◆ **Data Driven Decision Making (DDDM)**: using data to inform the decision-making process and validate a course of action before committing to it.
- ◆ “You can’t improve, what you can’t measure” (Peter Drucker)
- ◆ DDDM Approach
 1. Formulate Vision and Hypothesis
 2. Find, Collect, Organize Data
 3. Perform Data Analysis
 4. Draw Conclusions and Strategize
 5. Measure Success and Iterate
- ◆ Note: intuition, common sense, human creativity vital, esp. in step 1.

Intuitive Decision Making



- Relies on Gut Feeling (Subjective)
- Relies heavily on Previous Experience
- Best Guesses: Inherently Risky
- Correctiv & Unpredictable

Data Driven Decision Making



- Relies on Facts (Objective)
- Inspired by Previous Experiences
- Hypothesis Testing: Risk Averse
- Directive & Predictable

Five Stages towards a Data Driven Company Culture

5

Data-Driven

Data plays a central role in as many decisions as possible.

4

Data-Informed

Managers use data selectively to aid decision-making.

3

Data-Aware

Data is collected for monitoring, but not as basis for decisions.

2

Data Indifference

Company has no interest in whether data is collected or used.

1

Data Denial

Company has an active distrust of data and does not use it.

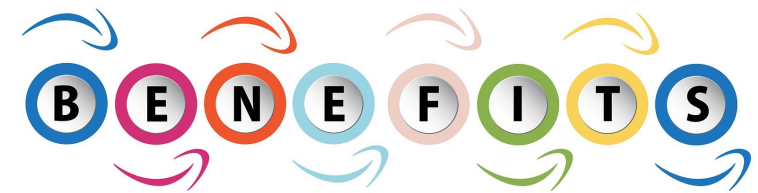
Benefits of DDDM

◆ Benefits of DDDM

- Make more confident, defensible decisions
- Guard against Biases
- Manage Risks
- Measurable Goals

◆ Financial Benefit for companies in Stages 4 & 5

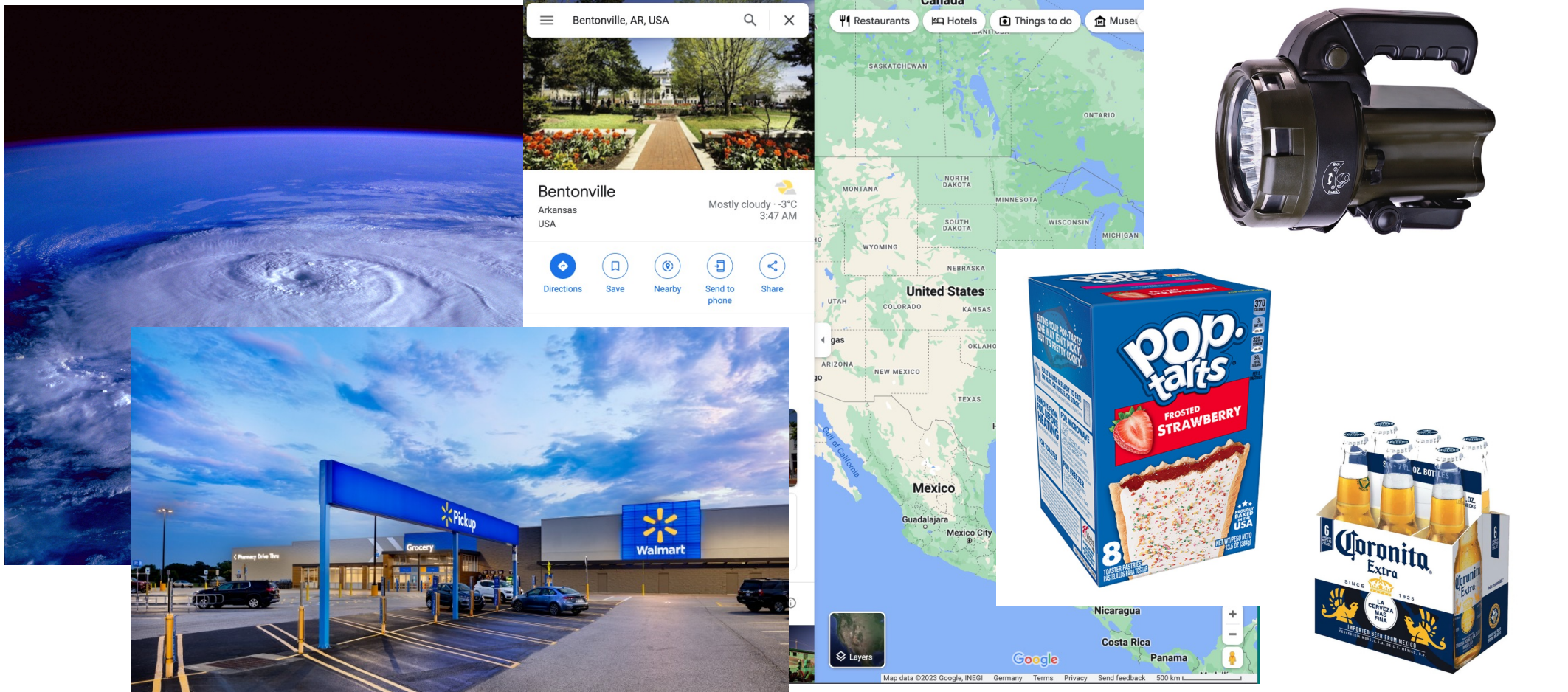
- 5-6% higher output and productivity⁽¹⁾



Sources:

(1) E. Brynjolfsson; Strength in Numbers: How Does Data-Driven Decisionmaking Affect Firm Performance?; 2011, SSRN Electronic Journal 1; DOI:10.2139/ssrn.1819486

Example: “What Wal-Mart Knows About Customers’ Habits”, 2004



Sources: <https://pixabay.com/de/photos/hurrikan-erde-satellit-verfolgung-92968/>, <https://corporate.walmart.com/about>, <https://www.google.com/maps>, <https://pixabay.com/de/photos/lamp-small-flashlight-licht-4862982/>, https://www.poptarts.com/en_US/products/favorites/pop-tarts-frosted-strawberry-product.html, <https://www.walmart.co.cr/6-pack-cerveza-corona-extra-botella-207ml/p>



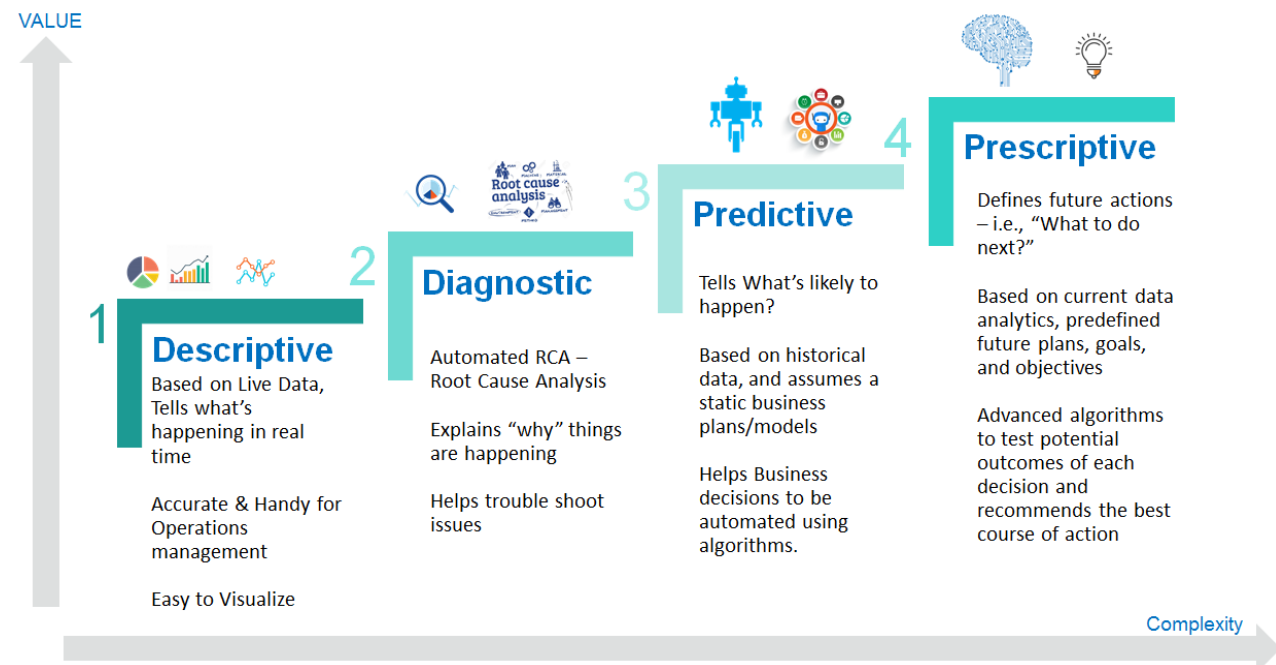
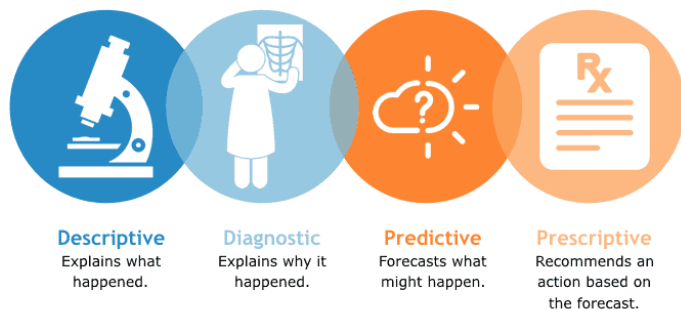
Example: “What Wal-Mart Knows About Customers’ Habits”, 2004

- ◆ “HURRICANE FRANCES was on its way, barrelling across the Caribbean, threatening a direct hit on Florida's Atlantic coast. Residents made for higher ground, but far away, in Bentonville, Ark., executives at Wal-Mart Stores decided that the situation offered a great opportunity for one of their newest data-driven weapons, something that the company calls predictive technology.
- ◆ A week ahead of the storm's landfall, Linda M. Dillman, Wal-Mart's chief information officer, pressed her staff to come up with forecasts based on what had happened when Hurricane Charley struck several weeks earlier. Backed by the trillions of bytes' worth of shopper history that is stored in Wal-Mart's computer network, she felt that the company could "start predicting what's going to happen, instead of waiting for it to happen," as she put it.
- ◆ The experts mined the data and found that the stores would indeed need certain products -- and not just the usual flashlights. "We didn't know in the past that strawberry Pop-Tarts increase in sales, like seven times their normal sales rate, ahead of a hurricane," Ms. Dillman said in a recent interview. "And the pre-hurricane top-selling item was beer."
- ◆ Thanks to those insights, trucks filled with toaster pastries and six-packs were soon speeding down Interstate 95 toward Wal-Marts in the path of Frances. Most of the products that were stocked for the storm sold quickly, the company said.
- ◆ Such knowledge, Wal-Mart has learned, is not only power. It is profit, too.”

Source: C. Hays, New York Times, 2004; <https://www.nytimes.com/2004/11/14/business/yourmoney/what-walmart-knows-about-customers-habits.html>

Descriptive – Diagnostic – Predictive – Prescriptive Data Analytics

- ♦ data analytics → answering questions & making decisions
- ♦ different types of questions → different types of data analytics
- ♦ Four primary types of data analytics:



Source: <https://medium.com/co-learning-lounge/types-of-data-analytics-descriptive-diagnostic-predictive-prescriptive-922654ce8f8f>

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Data Science Process

Business Problem vs. Data Science Problem

human psychology often plays a role in what the most optimal (and realistic) course of action is, e.g. 19 false alarm to every 20th being real may be cost efficient but will lead to people ignoring the alarm

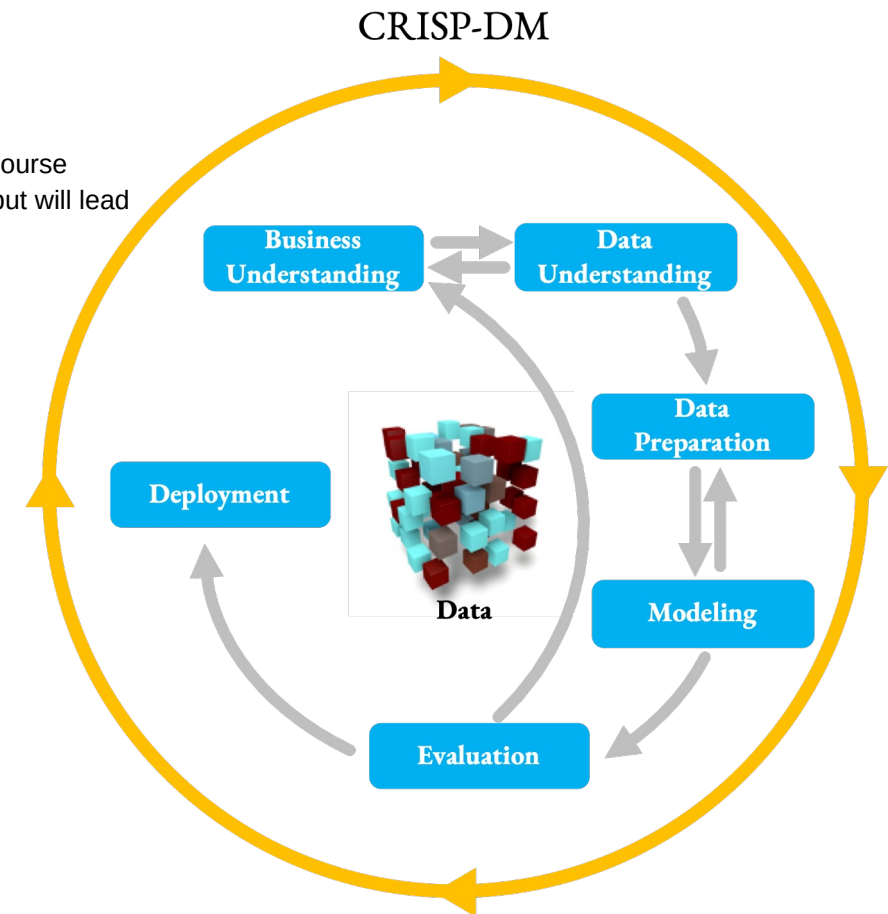
Data Acquisition, EDA, Visual Analysis

Data Cleaning, Feature Engineering

Modelling and Model Optimization

Model Evaluation

Deployment

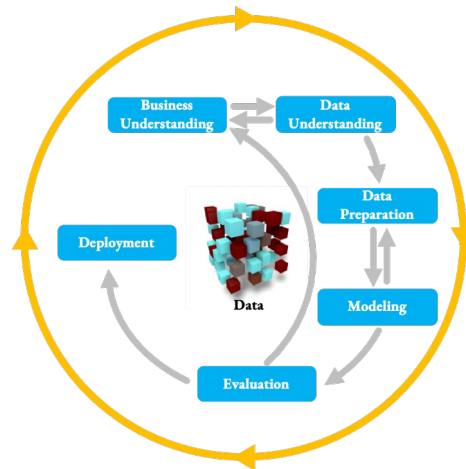


1) Business Understanding and 2) Data Understanding

Business Understanding

- Determine business objectives
- Assess situation
- Determine data mining goals
- Produce project plan

➔ Extremely critical phase



Data Understanding

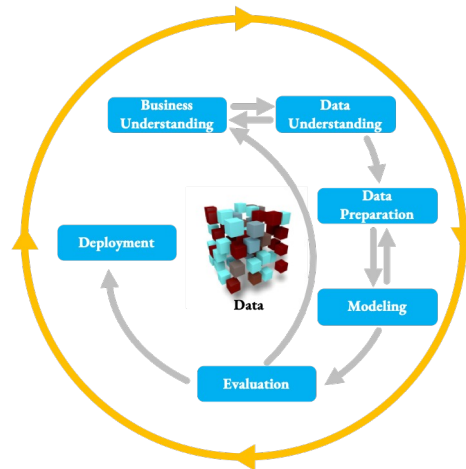
- Collect initial data
- Describe data
- Explore data
- Verify data quality

3) Data Preparation and 4) Modelling

Data Preparation

- Select data
- Clean data
- Construct data
- Integrate data
- Format data

➔ About 60%-80% of total effort!



Modelling

- Select modelling techniques
- Generate test design
- Build model
- Assess model

➔ Most exciting, but shortest phase!

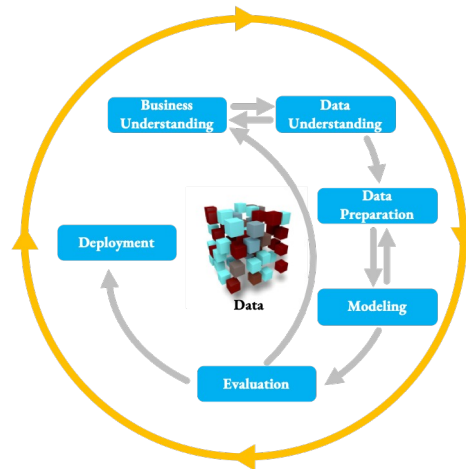
feature design, what data is actually important for the model and in what form
what can my model even do with the given data? e.g. system saves birth date but model
needs age, so age must be pre-calculated or model must be capable of difference calculation

5) Evaluation and 6) Deployment

Evaluation

- Evaluate results
- Review process
- Determine next steps

➔ Evaluation w.r.t. Business Problem



Deployment

- Plan deployment
- Plan monitoring and maintenance
- Produce final report
- Review project

➔ Not the end of the project
(Maintenance, Tuning, Retraining)

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Python

- ♦ Widely used high-level, general-purpose, interpreted, dynamic programming language
- ♦ Multiple programming paradigms: object-oriented, imperative, functional, procedural
- ♦ Dynamic type system
- ♦ Automatic memory management
- ♦ Large and comprehensive standard library.
- ♦ Python's name:
 - Derived from the Monty Python's Flying Circus
 - Common to use Monty Python references in example code (e.g. spam / eggs, instead of foo / bar)
 - Official Python documentation often contains various obscure Monty Python references!



Python History and Significance

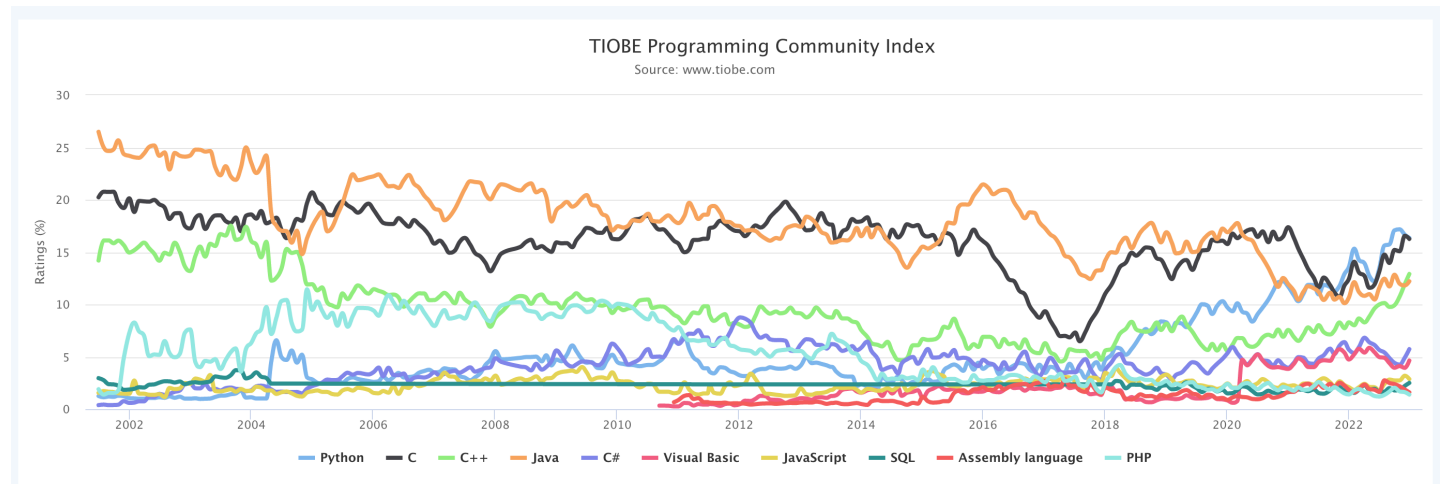
◆ Significance (Tiobe Index)

- Most popular language in Jan 2022 & 2023
- Language of the year in 2007 & 2010
- Esp. effective in scientific computing due to libraries (NumPy, SciPy, Pandas, Matplotlib, ...)

◆ History

- Conceived in the late 1980s.
- Python 2.0 released in 2000.
- Python 3.0 released in 2008: major, backwards-incompatible release.
- Python 3.14 released in Oct. 2025

Jan 2023	Jan 2022	Change	Programming Language	Ratings	Change
1	1		 Python	16.36%	+2.78%
2	2		 C	16.26%	+3.82%
3	4	▲	 C++	12.91%	+4.62%
4	3	▼	 Java	12.21%	+1.55%
5	5		 C#	5.73%	+0.05%
6	6		 Visual Basic	4.64%	-0.10%
7	7		 JavaScript	2.87%	+0.78%
8	9	▲	 SQL	2.50%	+0.70%



Source: <https://www.tiobe.com/tiobe-index/> (Jan 2023)

Jupyter Notebooks



- ◆ **Jupyter Notebook** (previously iPython Notebook)
 - browser-based graphical interface to the IPython shell (i.e. a web-app)
 - to create and share documents that contain live code, equations, visualizations and explanatory text.
 - Typical uses: data cleaning & transformation, statistical modelling, machine learning...

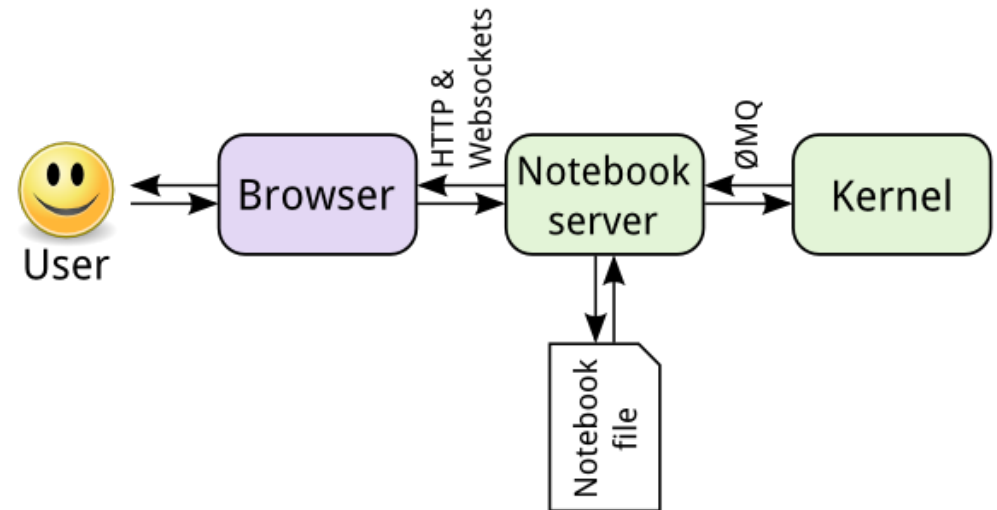
- ◆ **Notebook = JSON document =** ordered list of input/output cells
 - Most important cell types: code and markdown. Both types can be executed.
 - **code cells:** contain (in our case) Python 3 code.
 - When executed, the code is run and the result (or a syntax error!) is displayed
 - **markdown cells:** contain explanatory text
 - can be formatted in various way, including using a (subset) of HTML and CSS.
 - When executed, they are rendered "nicely".

- ◆ Further Reading: <https://jupyter.org/>

Architecture

◆ Notebook frontend (Browser)

- manages notebooks and runs the code
- stores code and output, together with markdown notes, in the notebook.
- upon save: notebook is sent from browser to notebook server and saved on disk as a JSON file (.ipynb file)



◆ Notebook server: responsible for saving and loading notebooks

- you can edit notebooks even if you don't have the kernel for that language, but cannot run the code

◆ Kernel: gets sent cells of code to execute

- doesn't know anything about the notebook document

Source: https://docs.jupyter.org/en/latest/_images/notebook_components.png

Linearity in Space and Time

- ♦ Python kernel running the code in the notebook is stateful
 - Potentially every time a cell is executed, the state in the kernel changes
- ♦ Cells are ordered (“written”) linearly in the notebook (“in space”)
- ♦ Cells be executed in any order. This generates a linear order of executions (“in time”).
 - You can go back and re-execute a cell as many times as you like.
 - They can even be changed and re-executed.
- ♦ Extremely useful for interactive development & extremely confusing, if not used carefully!

Absolute Rule For This Class

Every notebook **MUST** be linearly (spacewise) executable and give the correct result.

- ♦ Best practice: Use **idempotent cells!**

Key Takeaways

- ◆ What is **Data Science**
 - Esp. versus Machine Learning
- ◆ **DDDM**
 - What is Data Driven Decision Making
 - Why to think data analytically
- ◆ **Descriptive – Diagnostic – Predictive – Prescriptive Analytics**
- ◆ **CRISP-DM and it's 6 Phases**
- ◆ **Linearity in Space and Time**

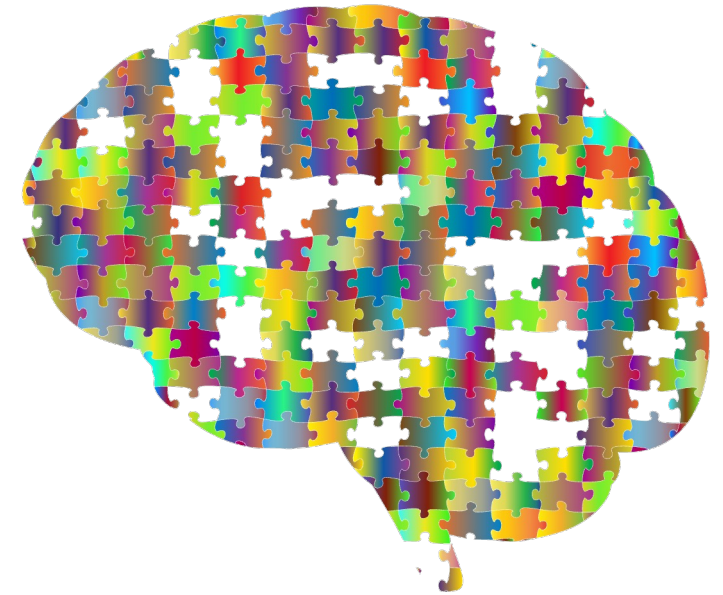


Image by Gordon Johnson on pixabay

Exercises

Exercise 1

- Set-up your system (git, docker, jupyter, ...).
- Execute the "Hello-World"-Notebook and check for errors.

Exercise 2

- Answer the questions and solve the exercises in notebook 2.

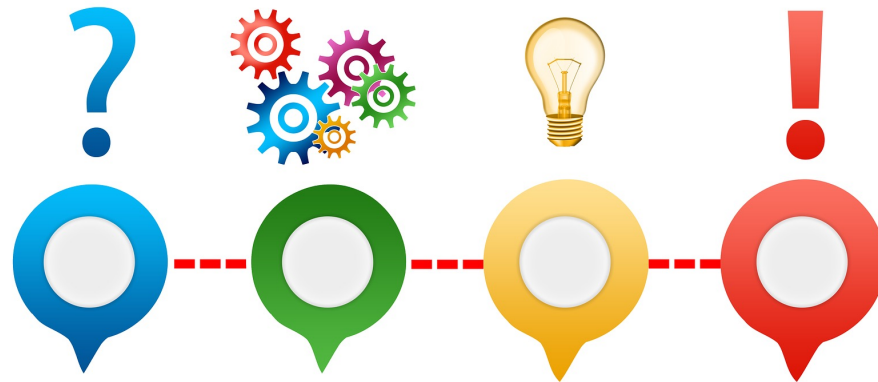


Photo by Gerd Altmann on Pixabay